

USER BEHAVIOR FUNNEL ANALYSIS FOR A SAAS PRODUCT

1. Introduction

The objective of this project was to analyse user behavior data for a SaaS platform and identify where users drop off during their journey from landing on the platform to upgrading to a paid plan. The analysis focused on funnel performance, user retention, acquisition channels, and cohort behavior.

The project involved cleaning raw event data, building a five-step conversion funnel, calculating conversion and drop-off rates, and performing segmentation and cohort analysis. Visual dashboards were also created to support data-driven insights.

2. Data Cleaning and Preparation

The dataset was cleaned before analysis to ensure accuracy and reproducibility.

The following preprocessing steps were performed:

- Removed duplicate rows and repeated user events
- Handled missing values in country and device columns
- Converted timestamp fields into datetime format
- Structured event-level data for funnel analysis

After cleaning, the dataset became suitable for conversion analysis and dashboard visualization.

3. Funnel Performance Analysis

A five-step conversion funnel was defined:

1. Landing
2. Sign-up
3. Onboarding
4. First Feature Use
5. Upgrade

The analysis showed that the number of users decreased at every stage of the funnel, indicating user drop-off during product usage.

Key observations:

- Most users successfully reached the Sign-up stage after Landing.
- A noticeable decline occurred during the Onboarding stage, suggesting friction in the onboarding experience.
- The largest drop-off was observed between First Feature Use and Upgrade, indicating that users are not converting to paid subscriptions efficiently.

This suggests that users may enjoy the product during the free trial but hesitate to upgrade due to pricing concerns, lack of perceived value, or insufficient engagement.

4. Cohort Analysis

Users were grouped by acquisition month to evaluate performance across cohorts.

The cohort analysis revealed that some monthly cohorts had stronger upgrade rates compared to others. This indicates differences in user quality, onboarding effectiveness, or acquisition campaign performance.

Some cohorts retained users more successfully and showed higher upgrade percentages, suggesting that specific campaigns or onboarding strategies were more effective during those months.

5. Segmentation Analysis

The analysis compared user behavior across acquisition channels:

- Organic
- Paid
- Referral

Findings showed that acquisition channels behaved differently in terms of user retention and paid conversions.

Paid users generally demonstrated stronger upgrade behavior, indicating higher intent and better targeting. Organic users showed larger drop-off in some funnel stages, while referral users displayed relatively stable engagement due to trust-based acquisition.

This suggests that marketing investment should prioritize channels with higher conversion potential.

6. Key Drop-Off Points

The major friction points identified were:

- Sign-up to Onboarding
- First Feature Use to Upgrade

Possible reasons include:

- Complex onboarding experience
 - Lack of product guidance
 - Weak free-trial engagement
 - Pricing concerns
 - Limited understanding of premium features
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7. Recommendations

Based on the analysis, the following recommendations are proposed:

1. Improve onboarding flow

Provide interactive tutorials and guided product walkthroughs.

2. Increase user engagement during trial

Send email reminders and product tips during free trial.

3. Improve upgrade conversion

Offer discounts, premium feature previews, or limited-time upgrade incentives.

4. Optimize acquisition channels

Focus marketing budget on channels with higher conversion rates.

5. Conduct A/B testing

Test different onboarding flows and pricing strategies to improve retention.

8. Conclusion

The SaaS funnel analysis successfully identified user behavior patterns, conversion bottlenecks, and acquisition performance. The findings indicate that improving onboarding and increasing engagement before the upgrade stage can significantly improve retention and paid subscription rates.

This analysis provides valuable business insights that can help the company optimize user experience and increase revenue growth.